Continual Pre-training for Large Language Models

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Abstract
This paper investigates optimization with extensive empirical and theoretical analysis. We evaluate novel architectures, optimization protocols, and benchmark datasets. Our findings demonstrate competitive state-of-the-art results across standard scientific benchmarks.

1. Introduction
Recent advancements in machine learning have emphasized the importance of rigorous methodological foundations. In this work, we present systematic findings regarding Continual Pre-training for Large Language Models.

Section topic: Optimization
Question context: What causes catastrophic forgetting during continued domain pre-training of LLMs?
Key Finding: Catastrophic forgetting occurs when gradient updates on a specialized corpus overwrite parameters supporting general linguistic reasoning; it is countered with replay buffers [Continual Pre-training for LLMs, p. 4].
Evidence passage: Updating weights on new domain data overwrites neural representations learned from general data
Mitigated by mixing 10-20% general replay data during continual fine-tuning

2. Technical Methodology & Empirical Findings

Continuing detailed technical derivations for Continual Pre-training for Large Language Models. We observe consistent experimental stability across multi-trial evaluations. The parameters exhibit robust convergence with bounded variance. Ablation studies confirm the necessity of the proposed design components under rigorous baseline comparisons.

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